

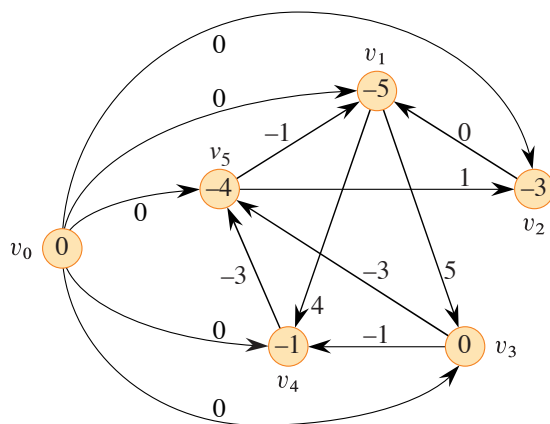


Figure 22.8 The constraint graph corresponding to the system (22.2)–(22.9) of difference constraints. The value of $\delta(v_0, v_i)$ appears in each vertex v_i . One feasible solution to the system is $x = (-5, -3, 0, -1, -4)$.

$$V = \{v_0, v_1, \dots, v_n\}$$

and

$$E = \{(v_i, v_j) : x_j - x_i \leq b_k \text{ is a constraint}\} \\ \cup \{(v_0, v_1), (v_0, v_2), (v_0, v_3), \dots, (v_0, v_n)\}.$$

The constraint graph includes the additional vertex v_0 , as we shall see shortly, to guarantee that the graph has some vertex that can reach all other vertices. Thus, the vertex set V consists of a vertex v_i for each unknown x_i , plus an additional vertex v_0 . The edge set E contains an edge for each difference constraint, plus an edge (v_0, v_i) for each unknown x_i . If $x_j - x_i \leq b_k$ is a difference constraint, then the weight of edge (v_i, v_j) is $w(v_i, v_j) = b_k$. The weight of each edge leaving v_0 is 0. Figure 22.8 shows the constraint graph for the system (22.2)–(22.9) of difference constraints.

The following theorem shows how to solve a system of difference constraints by finding shortest-path weights in the corresponding constraint graph.

Theorem 22.9

Given a system $Ax \leq b$ of difference constraints, let $G = (V, E)$ be the corresponding constraint graph. If G contains no negative-weight cycles, then

$$x = (\delta(v_0, v_1), \delta(v_0, v_2), \delta(v_0, v_3), \dots, \delta(v_0, v_n)) \quad (22.10)$$

is a feasible solution for the system. If G contains a negative-weight cycle, then there is no feasible solution for the system.

Proof We first show that if the constraint graph contains no negative-weight cycles, then equation (22.10) gives a feasible solution. Consider any edge $(v_i, v_j) \in E$. The triangle inequality implies that $\delta(v_0, v_j) \leq \delta(v_0, v_i) + w(v_i, v_j)$, which is equivalent to $\delta(v_0, v_j) - \delta(v_0, v_i) \leq w(v_i, v_j)$. Thus, letting $x_i = \delta(v_0, v_i)$ and $x_j = \delta(v_0, v_j)$ satisfies the difference constraint $x_j - x_i \leq w(v_i, v_j)$ that corresponds to edge (v_i, v_j) .

Now we show that if the constraint graph contains a negative-weight cycle, then the system of difference constraints has no feasible solution. Without loss of generality, let the negative-weight cycle be $c = \langle v_1, v_2, \dots, v_k \rangle$, where $v_1 = v_k$. (The vertex v_0 cannot be on cycle c , because it has no entering edges.) Cycle c corresponds to the following difference constraints:

$$\begin{aligned} x_2 - x_1 &\leq w(v_1, v_2), \\ x_3 - x_2 &\leq w(v_2, v_3), \\ &\vdots \\ x_{k-1} - x_{k-2} &\leq w(v_{k-2}, v_{k-1}), \\ x_k - x_{k-1} &\leq w(v_{k-1}, v_k). \end{aligned}$$

We'll assume that x has a solution satisfying each of these k inequalities and then derive a contradiction. The solution must also satisfy the inequality that results from summing the k inequalities together. In summing the left-hand sides, each unknown x_i is added in once and subtracted out once (remember that $v_1 = v_k$ implies $x_1 = x_k$), so that the left-hand side sums to 0. The right-hand side sums to the weight $w(c)$ of the cycle, giving $0 \leq w(c)$. But since c is a negative-weight cycle, $w(c) < 0$, and we obtain the contradiction that $0 \leq w(c) < 0$. ■

Solving systems of difference constraints

Theorem 22.9 suggests how to use the Bellman-Ford algorithm to solve a system of difference constraints. Because the constraint graph contains edges from the source vertex v_0 to all other vertices, any negative-weight cycle in the constraint graph is reachable from v_0 . If the Bellman-Ford algorithm returns TRUE, then the shortest-path weights give a feasible solution to the system. In Figure 22.8, for example, the shortest-path weights provide the feasible solution $x = (-5, -3, 0, -1, -4)$, and by Lemma 22.8, $x = (d - 5, d - 3, d, d - 1, d - 4)$ is also a feasible solution for any constant d . If the Bellman-Ford algorithm returns FALSE, there is no feasible solution to the system of difference constraints.

A system of difference constraints with m constraints on n unknowns produces a graph with $n + 1$ vertices and $n + m$ edges. Thus, the Bellman-Ford algorithm provides a way to solve the system in $O((n + 1)(n + m)) = O(n^2 + nm)$ time.

Exercise 22.4-5 asks you to modify the algorithm to run in $O(nm)$ time, even if m is much less than n .

Exercises

22.4-1

Find a feasible solution or determine that no feasible solution exists for the following system of difference constraints:

$$x_1 - x_2 \leq 1 ,$$

$$x_1 - x_4 \leq -4 ,$$

$$x_2 - x_3 \leq 2 ,$$

$$x_2 - x_5 \leq 7 ,$$

$$x_2 - x_6 \leq 5 ,$$

$$x_3 - x_6 \leq 10 ,$$

$$x_4 - x_2 \leq 2 ,$$

$$x_5 - x_1 \leq -1 ,$$

$$x_5 - x_4 \leq 3 ,$$

$$x_6 - x_3 \leq -8 .$$

22.4-2

Find a feasible solution or determine that no feasible solution exists for the following system of difference constraints:

$$x_1 - x_2 \leq 4 ,$$

$$x_1 - x_5 \leq 5 ,$$

$$x_2 - x_4 \leq -6 ,$$

$$x_3 - x_2 \leq 1 ,$$

$$x_4 - x_1 \leq 3 ,$$

$$x_4 - x_3 \leq 5 ,$$

$$x_4 - x_5 \leq 10 ,$$

$$x_5 - x_3 \leq -4 ,$$

$$x_5 - x_4 \leq -8 .$$

22.4-3

Can any shortest-path weight from the new vertex v_0 in a constraint graph be positive? Explain.

22.4-4

Express the single-pair shortest-path problem as a linear program.

22.4-5

Show how to modify the Bellman-Ford algorithm slightly so that when using it to solve a system of difference constraints with m inequalities on n unknowns, the running time is $O(nm)$.

22.4-6

Consider adding *equality constraints* of the form $x_i = x_j + b_k$ to a system of difference constraints. Show how to solve this variety of constraint system.

22.4-7

Show how to solve a system of difference constraints by a Bellman-Ford-like algorithm that runs on a constraint graph without the extra vertex v_0 .

★ 22.4-8

Let $Ax \leq b$ be a system of m difference constraints in n unknowns. Show that the Bellman-Ford algorithm, when run on the corresponding constraint graph, maximizes $\sum_{i=1}^n x_i$ subject to $Ax \leq b$ and $x_i \leq 0$ for all x_i .

★ 22.4-9

Show that the Bellman-Ford algorithm, when run on the constraint graph for a system $Ax \leq b$ of difference constraints, minimizes the quantity $(\max \{x_i\} - \min \{x_i\})$ subject to $Ax \leq b$. Explain how this fact might come in handy if the algorithm is used to schedule construction jobs.

22.4-10

Suppose that every row in the matrix A of a linear program $Ax \leq b$ corresponds to a difference constraint, a single-variable constraint of the form $x_i \leq b_k$, or a single-variable constraint of the form $-x_i \leq b_k$. Show how to adapt the Bellman-Ford algorithm to solve this variety of constraint system.

22.4-11

Give an efficient algorithm to solve a system $Ax \leq b$ of difference constraints when all of the elements of b are real-valued and all of the unknowns x_i must be integers.

★ 22.4-12

Give an efficient algorithm to solve a system $Ax \leq b$ of difference constraints when all of the elements of b are real-valued and a specified subset of some, but not necessarily all, of the unknowns x_i must be integers.

22.5 Proofs of shortest-paths properties

Throughout this chapter, our correctness arguments have relied on the triangle inequality, upper-bound property, no-path property, convergence property, path-relaxation property, and predecessor-subgraph property. We stated these properties without proof on page 611. In this section, we prove them.

The triangle inequality

In studying breadth-first search (Section 20.2), we proved as Lemma 20.1 a simple property of shortest distances in unweighted graphs. The triangle inequality generalizes the property to weighted graphs.

Lemma 22.10 (Triangle inequality)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$ and source vertex s . Then, for all edges $(u, v) \in E$,

$$\delta(s, v) \leq \delta(s, u) + w(u, v).$$

Proof Suppose that p is a shortest path from source s to vertex v . Then p has no more weight than any other path from s to v . Specifically, path p has no more weight than the particular path that takes a shortest path from source s to vertex u and then takes edge (u, v) .

Exercise 22.5-3 asks you to handle the case in which there is no shortest path from s to v . ■

Effects of relaxation on shortest-path estimates

The next group of lemmas describes how shortest-path estimates are affected by executing a sequence of relaxation steps on the edges of a weighted, directed graph that has been initialized by INITIALIZE-SINGLE-SOURCE.

Lemma 22.11 (Upper-bound property)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$. Let $s \in V$ be the source vertex, and let the graph be initialized by INITIALIZE-SINGLE-SOURCE(G, s). Then, $v.d \geq \delta(s, v)$ for all $v \in V$, and this invariant is maintained over any sequence of relaxation steps on the edges of G . Moreover, once $v.d$ achieves its lower bound $\delta(s, v)$, it never changes.

Proof We prove the invariant $v.d \geq \delta(s, v)$ for all vertices $v \in V$ by induction over the number of relaxation steps.

For the base case, $v.d \geq \delta(s, v)$ holds after initialization, since if $v.d = \infty$, then $v.d \geq \delta(s, v)$ for all $v \in V - \{s\}$, and since $s.d = 0 \geq \delta(s, s)$. (Note that $\delta(s, s) = -\infty$ if s is on a negative-weight cycle and that $\delta(s, s) = 0$ otherwise.)

For the inductive step, consider the relaxation of an edge (u, v) . By the inductive hypothesis, $x.d \geq \delta(s, x)$ for all $x \in V$ prior to the relaxation. The only d value that may change is $v.d$. If it changes, we have

$$\begin{aligned} v.d &= u.d + w(u, v) \\ &\geq \delta(s, u) + w(u, v) \quad (\text{by the inductive hypothesis}) \\ &\geq \delta(s, v) \quad (\text{by the triangle inequality}), \end{aligned}$$

and so the invariant is maintained.

The value of $v.d$ never changes once $v.d = \delta(s, v)$ because, having achieved its lower bound, $v.d$ cannot decrease since we have just shown that $v.d \geq \delta(s, v)$, and it cannot increase because relaxation steps do not increase d values. ■

Corollary 22.12 (No-path property)

Suppose that in a weighted, directed graph $G = (V, E)$ with weight function $w : E \rightarrow \mathbb{R}$, no path connects a source vertex $s \in V$ to a given vertex $v \in V$. Then, after the graph is initialized by INITIALIZE-SINGLE-SOURCE(G, s), we have $v.d = \delta(s, v) = \infty$, and this equation is maintained as an invariant over any sequence of relaxation steps on the edges of G .

Proof By the upper-bound property, we always have $\infty = \delta(s, v) \leq v.d$, and thus $v.d = \infty = \delta(s, v)$. ■

Lemma 22.13

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$, and let $(u, v) \in E$. Then, immediately after edge (u, v) is relaxed by a call of RELAX(u, v, w), we have $v.d \leq u.d + w(u, v)$.

Proof If, just prior to relaxing edge (u, v) , we have $v.d > u.d + w(u, v)$, then $v.d = u.d + w(u, v)$ afterward. If, instead, $v.d \leq u.d + w(u, v)$ just before the relaxation, then neither $u.d$ nor $v.d$ changes, and so $v.d \leq u.d + w(u, v)$ afterward. ■

Lemma 22.14 (Convergence property)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$, let $s \in V$ be a source vertex, and let $s \rightsquigarrow u \rightarrow v$ be a shortest path in G for some vertices $u, v \in V$. Suppose that G is initialized by INITIALIZE-SINGLE-SOURCE(G, s) and then a sequence of relaxation steps that includes the call

$\text{RELAX}(u, v, w)$ is executed on the edges of G . If $u.d = \delta(s, u)$ at any time prior to the call, then $v.d = \delta(s, v)$ at all times after the call.

Proof By the upper-bound property, if $u.d = \delta(s, u)$ at some point prior to relaxing edge (u, v) , then this equation holds thereafter. In particular, after edge (u, v) is relaxed, we have

$$\begin{aligned} v.d &\leq u.d + w(u, v) && \text{(by Lemma 22.13)} \\ &= \delta(s, u) + w(u, v) \\ &= \delta(s, v) && \text{(by Lemma 22.1 on page 606).} \end{aligned}$$

The upper-bound property gives $v.d \geq \delta(s, v)$, from which we conclude that $v.d = \delta(s, v)$, and this equation is maintained thereafter. ■

Lemma 22.15 (Path-relaxation property)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$, and let $s \in V$ be a source vertex. Consider any shortest path $p = \langle v_0, v_1, \dots, v_k \rangle$ from $s = v_0$ to v_k . If G is initialized by $\text{INITIALIZE-SINGLE-SOURCE}(G, s)$ and then a sequence of relaxation steps occurs that includes, in order, relaxing the edges $(v_0, v_1), (v_1, v_2), \dots, (v_{k-1}, v_k)$, then $v_k.d = \delta(s, v_k)$ after these relaxations and at all times afterward. This property holds no matter what other edge relaxations occur, including relaxations that are intermixed with relaxations of the edges of p .

Proof We show by induction that after the i th edge of path p is relaxed, we have $v_i.d = \delta(s, v_i)$. For the base case, $i = 0$, and before any edges of p have been relaxed, we have from the initialization that $v_0.d = s.d = 0 = \delta(s, s)$. By the upper-bound property, the value of $s.d$ never changes after initialization.

For the inductive step, assume that $v_{i-1}.d = \delta(s, v_{i-1})$. What happens when edge (v_{i-1}, v_i) is relaxed? By the convergence property, after this relaxation, we have $v_i.d = \delta(s, v_i)$, and this equation is maintained at all times thereafter. ■

Relaxation and shortest-paths trees

We now show that once a sequence of relaxations has caused the shortest-path estimates to converge to shortest-path weights, the predecessor subgraph G_π induced by the resulting π values is a shortest-paths tree for G . We start with the following lemma, which shows that the predecessor subgraph always forms a rooted tree whose root is the source.

Lemma 22.16

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$, let $s \in V$ be a source vertex, and assume that G contains no negative-weight

cycles that are reachable from s . Then, after the graph is initialized by INITIALIZE-SINGLE-SOURCE(G, s), the predecessor subgraph G_π forms a rooted tree with root s , and any sequence of relaxation steps on edges of G maintains this property as an invariant.

Proof Initially, the only vertex in G_π is the source vertex, and the lemma is trivially true. Consider a predecessor subgraph G_π that arises after a sequence of relaxation steps. We first prove that G_π is acyclic. Suppose for the sake of contradiction that some relaxation step creates a cycle in the graph G_π . Let the cycle be $c = \langle v_0, v_1, \dots, v_k \rangle$, where $v_k = v_0$. Then, $v_i.\pi = v_{i-1}$ for $i = 1, 2, \dots, k$ and, without loss of generality, assume that relaxing edge (v_{k-1}, v_k) created the cycle in G_π .

We claim that all vertices on cycle c are reachable from the source vertex s . Why? Each vertex on c has a non-NIL predecessor, and so each vertex on c was assigned a finite shortest-path estimate when it was assigned its non-NIL π value. By the upper-bound property, each vertex on cycle c has a finite shortest-path weight, which means that it is reachable from s .

We'll examine the shortest-path estimates on cycle c immediately before the call RELAX(v_{k-1}, v_k, w) and show that c is a negative-weight cycle, thereby contradicting the assumption that G contains no negative-weight cycles that are reachable from the source. Just before the call, we have $v_i.\pi = v_{i-1}$ for $i = 1, 2, \dots, k-1$. Thus, for $i = 1, 2, \dots, k-1$, the last update to $v_i.d$ was by the assignment $v_i.d = v_{i-1}.d + w(v_{i-1}, v_i)$. If $v_{i-1}.d$ changed since then, it decreased. Therefore, just before the call RELAX(v_{k-1}, v_k, w), we have

$$v_i.d \geq v_{i-1}.d + w(v_{i-1}, v_i) \quad \text{for all } i = 1, 2, \dots, k-1. \quad (22.11)$$

Because $v_k.\pi$ is changed by the call RELAX(v_{k-1}, v_k, w), immediately beforehand we also have the strict inequality

$$v_k.d > v_{k-1}.d + w(v_{k-1}, v_k).$$

Summing this strict inequality with the $k-1$ inequalities (22.11), we obtain the sum of the shortest-path estimates around cycle c :

$$\begin{aligned} \sum_{i=1}^k v_i.d &> \sum_{i=1}^k (v_{i-1}.d + w(v_{i-1}, v_i)) \\ &= \sum_{i=1}^k v_{i-1}.d + \sum_{i=1}^k w(v_{i-1}, v_i). \end{aligned}$$

But

$$\sum_{i=1}^k v_i.d = \sum_{i=1}^k v_{i-1}.d,$$

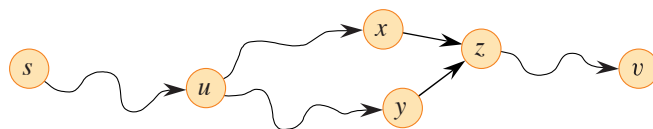


Figure 22.9 Showing that a simple path in G_π from source vertex s to vertex v is unique. If G_π contains two paths p_1 ($s \rightsquigarrow u \rightsquigarrow x \rightarrow z \rightsquigarrow v$) and p_2 ($s \rightsquigarrow u \rightsquigarrow y \rightarrow z \rightsquigarrow v$), where $x \neq y$, then $z.\pi = x$ and $z.\pi = y$, a contradiction.

since each vertex in the cycle c appears exactly once in each summation. This equation implies

$$0 > \sum_{i=1}^k w(v_{i-1}, v_i) .$$

Thus, the sum of weights around the cycle c is negative, which provides the desired contradiction.

We have now proven that G_π is a directed, acyclic graph. To show that it forms a rooted tree with root s , it suffices (see Exercise B.5-2 on page 1175) to prove that for each vertex $v \in V_\pi$, there is a unique simple path from s to v in G_π .

The vertices in V_π are those with non-NIL π values, plus s . Exercise 22.5-6 asks you to prove that a path from s exists to each vertex in V_π .

To complete the proof of the lemma, we now show that for any vertex $v \in V_\pi$, the graph G_π contains at most one simple path from s to v . Suppose otherwise. That is, suppose that, as Figure 22.9 illustrates, G_π contains two simple paths from s to some vertex v : p_1 , which we decompose into $s \rightsquigarrow u \rightsquigarrow x \rightarrow z \rightsquigarrow v$, and p_2 , which we decompose into $s \rightsquigarrow u \rightsquigarrow y \rightarrow z \rightsquigarrow v$, where $x \neq y$ (though u could be s and z could be v). But then, $z.\pi = x$ and $z.\pi = y$, which implies the contradiction that $x = y$. We conclude that G_π contains a unique simple path from s to v , and thus G_π forms a rooted tree with root s . ■

We can now show that if all vertices have been assigned their true shortest-path weights after a sequence of relaxation steps, then the predecessor subgraph G_π is a shortest-paths tree.

Lemma 22.17 (Predecessor-subgraph property)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$, let $s \in V$ be a source vertex, and assume that G contains no negative-weight cycles that are reachable from s . Then, after a call to INITIALIZE-SINGLE-SOURCE(G, s) followed by any sequence of relaxation steps on edges of G that produces $v.d = \delta(s, v)$ for all $v \in V$, the predecessor subgraph G_π is a shortest-paths tree rooted at s .

Proof We must prove that the three properties of shortest-paths trees given on page 608 hold for G_π . To show the first property, we must show that V_π is the set of vertices reachable from s . By definition, a shortest-path weight $\delta(s, v)$ is finite if and only if v is reachable from s , and thus the vertices that are reachable from s are exactly those with finite d values. But a vertex $v \in V - \{s\}$ has been assigned a finite value for $v.d$ if and only if $v.\pi \neq \text{NIL}$, since both assignments occur in RELAX. Thus, the vertices in V_π are exactly those reachable from s .

The second property, that G_π forms a rooted tree with root s , follows directly from Lemma 22.16.

It remains, therefore, to prove the last property of shortest-paths trees: for each vertex $v \in V_\pi$, the unique simple path $s \xrightarrow{p} v$ in G_π is a shortest path from s to v in G . Let $p = \langle v_0, v_1, \dots, v_k \rangle$, where $v_0 = s$ and $v_k = v$. Consider an edge (v_{i-1}, v_i) in path p . Because this edge belongs to G_π , the last relaxation that changed $v_i.d$ must have been of this edge. After that relaxation, we had $v_i.d = v_{i-1}.d + w(v_{i-1}, v_i)$. Subsequently, an edge entering v_{i-1} could have been relaxed, causing $v_{i-1}.d$ to decrease further, but without changing $v_i.d$. Therefore, we have $v_i.d \geq v_{i-1}.d + w(v_{i-1}, v_i)$. Thus, for $i = 1, 2, \dots, k$, we have both $v_i.d = \delta(s, v_i)$ and $v_i.d \geq v_{i-1}.d + w(v_{i-1}, v_i)$, which together imply $w(v_{i-1}, v_i) \leq \delta(s, v_i) - \delta(s, v_{i-1})$. Summing the weights along path p yields

$$\begin{aligned}
 w(p) &= \sum_{i=1}^k w(v_{i-1}, v_i) \\
 &\leq \sum_{i=1}^k (\delta(s, v_i) - \delta(s, v_{i-1})) \\
 &= \delta(s, v_k) - \delta(s, v_0) \quad (\text{because the sum telescopes}) \\
 &= \delta(s, v_k) \quad (\text{because } \delta(s, v_0) = \delta(s, s) = 0).
 \end{aligned}$$

Thus, we have $w(p) \leq \delta(s, v_k)$. Since $\delta(s, v_k)$ is a lower bound on the weight of any path from s to v_k , we conclude that $w(p) = \delta(s, v_k)$, and p is a shortest path from s to $v = v_k$. ■

Exercises

22.5-1

Give two shortest-paths trees for the directed graph of Figure 22.2 on page 609 other than the two shown.

22.5-2

Give an example of a weighted, directed graph $G = (V, E)$ with weight function $w : E \rightarrow \mathbb{R}$ and source vertex s such that G satisfies the following property: For